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Deep Learning (285652)

Beyond Clean Data

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1 Abstract

Robustness under distribution shift is essential for image classifiers, because clean benchmark accuracy alone does not show whether predictions remain stable when inputs change in realistic ways. This report compares a Multi-Layer Perceptron (MLP) and a Convolutional Neural Network (CNN) on MNIST under systematic perturbations. Both models were trained only on clean MNIST and evaluated on rotated, translated, noisy, blurred, eroded, inverted, and hand-drawn digits. On the clean test set, both achieved strong baseline performance (MLP: 98.03%, CNN: 99.11%). However, their robustness profiles differed substantially. The CNN was clearly more robust to translation, blur, erosion, and inversion; for example, at a translation of 3 pixels it still reached 72.22% accuracy, whereas the MLP dropped to 21.27%. This supports the view that convolutional architectures benefit from local receptive fields and weight sharing, which reduce reliance on absolute pixel position. Yet the CNN advantage was not universal: under severe Gaussian noise, the MLP slightly outperformed the CNN. Both models also failed under strong distribution shift, especially on inverted and hand-drawn digits. Overall, the results show that architectural bias shapes robustness, but neither architecture provides general robustness on its own.

2 Introduction

High accuracy on a benchmark test set does not guarantee robustness under distribution shift. In practice, images may be shifted, rotated, blurred, noisy, or written in styles that differ from the training distribution. Robustness therefore asks whether a model has learned task-relevant structure or merely exploits regularities specific to clean benchmark data.

MNIST remains a useful testbed for studying this question [2]. Both MLPs and CNNs achieve very high clean accuracy, so robustness differences cannot be explained simply by one model failing at the basic task. In addition, MNIST allows perturbations to be applied in a controlled and interpretable way, making it easier to connect performance drops to properties of the learned representation.

The comparison between MLPs and CNNs is especially meaningful because the two architectures encode different assumptions about images. An MLP treats the input as a flattened pixel vector and has no explicit notion of locality or feature reuse across space. A CNN instead applies shared filters to local neighborhoods and aggregates them through pooling, introducing inductive biases toward local feature detection and partial translation tolerance [1]. We therefore expect CNNs to be more robust when digit identity is preserved but position or local appearance changes, whereas MLPs should be more sensitive to perturbations that disrupt fixed pixel arrangements.

At the same time, prior work shows that convolution alone does not guarantee invariance to simple transformations [3, 4]. The relevant question is therefore not whether the CNN is

robust in an absolute sense, but how the failure modes of the two models differ and what those differences reveal about the representations they learn.

This report addresses the following research question:

How do MLP and CNN differ in their robustness to systematic image perturbations on MNIST, and which types of perturbation cause which model to fail?

3 Methods

3.1 Dataset

The experiments use MNIST, a dataset of 28×28 grayscale images of handwritten digits [2]. Following the notebook template, the original training set was split into 50,000 training images and 10,000 validation images using a fixed random seed. The official MNIST test set of 10,000 images was used for baseline and perturbation evaluation. In addition, a small hand-drawn set was used to probe generalization to a shifted writing style.

3.2 Models

Two baseline architectures were trained from scratch on clean MNIST only. The MLP takes the flattened 784-dimensional image as input and uses two fully connected hidden layers with 256 units each and ReLU activations. The CNN consists of three convolutional blocks with 16, 32, and 64 channels, using 3×3 kernels, ReLU activation, and max pooling, followed by a fully connected classifier with one hidden layer of 128 units.

These models were intentionally kept simple to make the comparison interpretable. The goal was not to maximize performance, but to compare how a fully connected and a convolutional architecture respond to the same controlled shifts.

3.3 Training Setup

Both models were trained with cross-entropy loss and the Adam optimizer (learning rate 0.001), using batch size 128 for 20 epochs. Inputs were normalized with the standard MNIST mean (0.1307) and standard deviation (0.3081). No data augmentation was used during training so that robustness differences would primarily reflect architectural bias rather than exposure to transformed data.

3.4 Perturbation Design

The test images were systematically modified using seven perturbation types. Rotation was applied at angles from 5° to 180° ; translation shifted the digit by 1 to 5 pixels; Gaussian noise added zero-mean noise with standard deviation 0.1 to 1.0; Gaussian blur used increasing

blur strengths; erosion progressively thinned the strokes; inversion reversed foreground and background intensities; and a hand-drawn set introduced a non-programmatic style shift.

These perturbations were chosen to test different assumptions. Rotation and translation probe geometric invariance, noise and blur test tolerance to corrupted local evidence, erosion probes dependence on stroke thickness and continuity, inversion tests sensitivity to foreground-background polarity, and hand-drawn digits assess whether the models learned a transferable concept of digit identity.

3.5 Hand-drawn Data

Hand-drawn images were loaded from a separate folder, resized to 28×28 , converted to grayscale, and inverted where necessary so that the polarity matched MNIST. The same normalization as for MNIST was then applied. In the available notebook output, the final hand-drawn set contained 30 images, three per class.

3.6 Evaluation Protocol

The primary metric was classification accuracy. For each programmatic perturbation, both models were evaluated across multiple intensity levels, allowing accuracy-degradation curves to be plotted. To analyze failure patterns rather than only aggregate performance, the notebook also generated per-class comparisons, confusion matrices, and qualitative failure examples for selected perturbations.

4 Results

4.1 Baseline Performance

Both models exceeded the assignment target of 98% test accuracy on clean MNIST, with the CNN performing slightly better.

Model	Accuracy (%)
MLP	98.03
CNN	99.11

Table 1: Baseline accuracy on the clean MNIST test set.

4.2 Robustness Trends Across Perturbations

Figure 1 shows accuracy at the strongest perturbation intensity. Overall, the CNN is more robust than the MLP on most perturbations, especially when local structure is preserved but shifted or moderately degraded.

The clearest advantage appears under translation: even small shifts sharply reduce MLP accuracy, whereas the CNN degrades more gradually, though it is not fully translation-invariant at higher shifts.

A similar pattern holds for rotation: the CNN consistently outperforms the MLP, but both models fail at extreme angles, indicating no learned rotation invariance from clean MNIST.

For blur and erosion, the CNN again performs better, likely because it can still exploit partial local stroke structure after moderate degradation. Under very strong blur or erosion, however, both models collapse.

Inversion shows a strong contrast: the MLP nearly fails, while the CNN retains moderate performance, suggesting greater robustness to polarity changes.

Gaussian noise is the main exception. While the CNN performs slightly better at low levels, the MLP surpasses it at higher noise, indicating the CNN’s reliance on coherent local features.

Finally, both models generalize poorly to hand-drawn data (MLP: 10%, CNN: 10%), but this result should be interpreted cautiously because the set contains only thirty images.

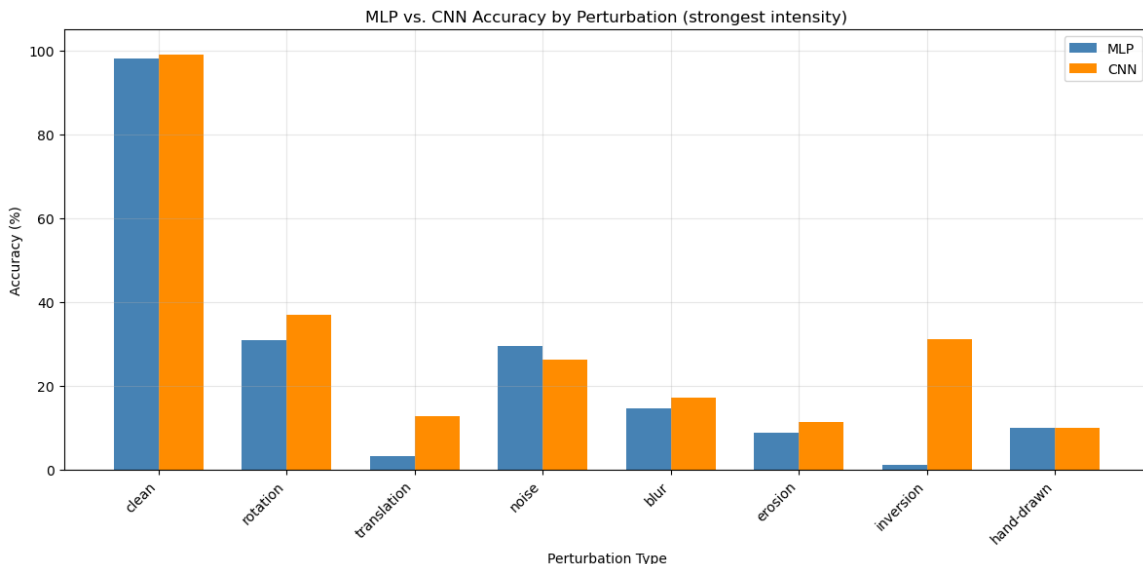


Figure 1: Accuracy of MLP and CNN under the strongest tested intensity for each perturbation type

Overall, the results show a clear robustness ranking: the CNN is consistently stronger for spatial shifts and structural degradation, while neither model is robust in a general sense.

4.3 Failure Patterns

The confusion-matrix analysis shows that the errors are systematic rather than random. Under translation, the MLP exhibits strong class collapse. At a shift of 3 pixels, many digits are reassigned to a small number of classes, most notably $1 \rightarrow 7$; at stronger shifts additional collapses such as $4 \rightarrow 2$ and $9 \rightarrow 2$ become dominant. This is consistent with a model that relies on fixed pixel locations: once a digit is displaced, the activation pattern no longer matches the spatial template learned during training.

The CNN shows a different translation failure mode. At moderate shifts it preserves much of the original class structure, explaining its large advantage over the MLP. At stronger shifts, however, its confusions also become systematic, especially $9 \rightarrow 6$, $6 \rightarrow 1$, and $4 \rightarrow 2$. This suggests that the CNN is not generally translation-invariant; it remains robust only while local filters and pooling can still recover familiar stroke configurations.

Rotation produces another interpretable pattern. For both models, the strongest confusions occur when rotation changes the perceived identity of a digit, such as $6 \leftrightarrow 9$ at 180° . At 90° , both models also show strong reassignment of $1 \rightarrow 7$. These errors are semantically meaningful: the models often map transformed digits to visually similar alternatives rather than failing randomly.

Blur and erosion lead to a third failure type. As fine stroke details disappear, the confusion matrices become more concentrated around digits with similar global shape. In the MLP, strong blur causes several classes to collapse toward 8, indicating that once edges and local separations are lost, many digits are reduced to a closed, blob-like pattern. The CNN resists this effect longer, but under severe structural degradation it also loses the topology required for reliable discrimination.

Overall, the confusion matrices reinforce the main interpretation of the accuracy results: the MLP fails mainly when perturbations violate position-specific pixel templates, whereas the CNN fails when perturbations destroy or strongly distort the local structures on which convolutional feature extraction depends.

5 Discussion

The results support the central hypothesis of the assignment: MLPs and CNNs fail for different reasons because they encode different assumptions about image structure. The CNN is more robust when perturbations preserve local patterns but change their exact placement or moderately degrade their appearance, which is most visible for translation, blur, and erosion. Local receptive fields, weight sharing, and pooling make the CNN less dependent on absolute coordinates and better able to reuse stroke features across nearby positions.

The MLP shows the opposite profile. Its sharp degradation under translation and inversion suggests that it relies primarily on position-specific and polarity-specific pixel correlations rather than on stable shape abstractions. The confusion matrices sharpen this interpretation: translation mainly breaks the MLP through class collapse into a few dominant wrong classes, while the CNN tends to fail only once the transformed digit begins to resemble another class or loses the local stroke fragments required for recognition. In this sense, the two models appear to learn different representations: the MLP behaves like a detector of fragile global pixel templates, whereas the CNN behaves more like a detector of reusable local stroke patterns.

At the same time, the experiments also show the limits of convolutional inductive bias. The CNN was better than the MLP on most perturbations, but it was not robust in an absolute

sense. It still degraded heavily under strong rotation, extreme erosion, inversion, and hand-drawn style shift. The noise crossover is especially informative because it shows that CNNs are not uniformly more robust; when noise destroys the local evidence on which convolution depends, the architectural advantage can disappear or even reverse.

Optional extension: a more robust CNN? To test whether robustness could be improved by architecture alone, a modified CNN (`CNN_v2`) was trained with larger early kernels, average pooling, and global average pooling. The result was mixed. On clean MNIST, `CNN_v2` reached 99.03%, essentially matching the original CNN (99.11%). It improved dramatically on translation by 3 pixels, increasing accuracy from 72.22% to 97.68%. However, it performed worse on most other perturbations, including rotation (47.66% vs. 60.90% at 45°), erosion (80.42% vs. 87.55% at 0.8), inversion (16.12% vs. 31.05%), hand-drawn digits (10.00% vs. 10.00%), and especially Gaussian noise (10.52% vs. 71.32% at noise level 0.5). This trade-off suggests that architectural changes can strengthen one invariance while weakening sensitivity to other useful cues.

The results therefore answer the research question directly: CNNs are more robust than MLPs mainly when task-relevant local structure remains available, whereas MLPs fail earlier when perturbations disrupt the exact pixel layouts seen during training.

Limitations. This study uses only MNIST, whose visual complexity is low. The architectures are simple baselines rather than strong modern models. No augmentation was used during training, so the results describe naive robustness rather than best achievable robustness. In addition, the available hand-drawn evaluation set contains only 30 images, which is useful qualitatively but too small for strong quantitative conclusions.

Implications and future work. The results suggest that architecture matters, but architecture alone is insufficient. Future work should test whether targeted augmentation or robustness-oriented training can improve performance without introducing the trade-offs observed in `CNN_v2`. Extending the analysis to larger and more varied handwritten datasets would also clarify which conclusions generalize beyond MNIST.

References

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